Microlocal Geometry of Étale Sheaves

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1 Introduction

k: a (perfect) field of characteristic $p > 0$.

Sch: the category of schemes of finite type over k.

Λ: a finite local ring with residue characteristic $\ell \neq p$, or $\mathbb{Q}_\ell$ or a finite extension of $\mathbb{Q}_\ell$.

For $X \in \mathsf{Sch}$, write $D(X) = D_{\mathrm{ctf}}(X, \Lambda)$.

Singular locus

For $X \xrightarrow{f} Y$, write

$$K_{X/Y} = f^!\Lambda \qquad \text{(in the derived sense)}, \qquad D_{X/Y}(-) = R\mathcal{H}om(-, K_{X/Y}).$$

Let $\mathcal{F} \in D(X)$. We say that $(f, \mathcal{F})$ is smooth, that f is $\mathcal{F}$-smooth ($\mathcal{F}$-ULA), or that $\mathcal{F}$ is f-smooth (f-ULA). By the criterion of Lu–Zheng, this is equivalent to the canonical map on $X \times_Y X$ being an isomorphism:

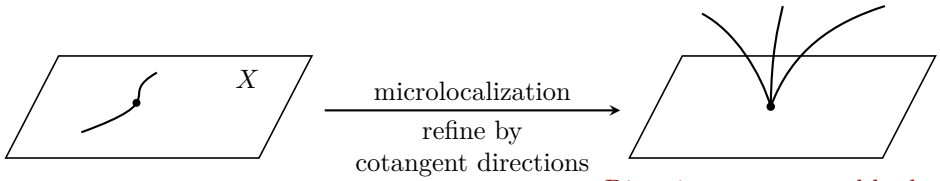
$$\mathcal{F} \boxtimes_Y D_{X/Y}(\mathcal{F}) \xrightarrow{\sim} R\mathcal{H}om\big(\mathrm{pr}_2^*\mathcal{F},\, \mathrm{pr}_1^!\mathcal{F}\big).$$

Thus, for $(f, \mathcal{F})$, we can talk about its singular locus.

Examples

1. If $Y = \operatorname{Spec} k$, then $(f, \mathcal{F})$ is smooth for every $\mathcal{F}$.
2. If $\mathcal{F} = \Lambda$ and f is smooth, then $(f, \mathcal{F})$ is smooth. The converse does not hold (consider a radicial map).
3. $(X \xrightarrow{\mathrm{id}} X, \mathcal{F})$ is smooth if and only if $\mathcal{F}$ is locally constant on X.
4. When Y is a curve, the vanishing cycles functor $R\Phi_f(\mathcal{F})$ provides a natural tool for detecting the singularities of $(f, \mathcal{F})$.

Idea. Consider $(X \xrightarrow{\mathrm{id}} X, \mathcal{F})$. A sheaf can have singularities not only at a point $x \in X$, but also in certain directions $\xi \in T_x^*X$.



Singular locus: where $\mathcal{F}$ is not locally constant

Directions represented by local functions $f : X \to \mathbf{A}^1$, detected by $R\Phi_f(\mathcal{F})$

Singular support. The singular support $\mathrm{SS}(\mathcal{F}) \subseteq T^*X$ is a closed conical subset. If $(x, \xi) \in \mathrm{SS}(\mathcal{F})$, the sheaf $\mathcal{F}$ may be singular at x in the direction ξ. In terms of local test functions, we write

$$\mathrm{SS}(\mathcal{F}) = \overline{\left\{ (x, df_x) \;\middle|\; \begin{array}{c} f \text{ a local function at } x, \\ R\Phi_f(\mathcal{F})_x \neq 0 \end{array} \right\}} \subseteq T^*X.$$

Beilinson. $\dim \mathrm{SS}(\mathcal{F}) = \dim X$.

Saito's characteristic cycle

$$\mathrm{CC}(\mathcal{F}) = \sum_{C_i \text{ irred. component of } \mathrm{SS}(\mathcal{F})} m_i [C_i]$$

is a quantitative enhancement of the singular support, in the sense that

$$-\mathrm{dimtot} R\Phi_f(\mathcal{F})_x = \big\langle \mathrm{CC}(\mathcal{F}), df \big\rangle_{T^*X, x}$$

when x is an isolated characteristic point of the local function f relative to $\mathcal{F}$.

Based on Kashiwara–Schapira's work in the transcendental setting over $\mathbb{C}$, there is an open question in ramification theory (I learned it from a talk given by Beilinson in 2015):

> Can we lift $\mathcal{F}$ to an object on T^*X whose support is $\mathrm{SS}(\mathcal{F})$ and whose characteristic class is $\mathrm{CC}(\mathcal{F})$?

Around 2004, Abbes and Saito used log blow-ups to kill wild ramification. At the time, they thought that Kashiwara–Schapira's construction might not work for ℓ-adic sheaves because of wild ramification: the concern was that specialization would lose the wild ramification.

> Does wild ramification really matter?

Two observations.

1. *Laumon's local Fourier transform.* Roughly,

$$\mathrm{dimtot} R\Phi = \dim R\Psi\big(- \otimes \mathcal{L}_\psi\big).$$

2. *Yang–Zhao's NA class.* The cohomological conductor formula takes the form

$$-a_y(\mathcal{F}) = \text{NA class} \qquad \text{(localized cohomological characteristic class).} \qquad (*)$$

   The left-hand side is

$$-\big\{\dim \mathcal{F}_{\bar{\eta}} - \dim \mathcal{F}_y + \mathrm{Sw}(\mathcal{F}_{\bar{\eta}_y})\big\}.$$

   The class on the right is purely categorical.

From $(*)$, I guessed that an appropriate form of microlocalization might exist: perhaps one could replace specialization by the NA class.

Saito calculated some examples: specialization itself works! (Just consider Laumon's formula.)

Theorem. For ℓ-adic sheaves, microlocalization exists.

2 Construction of microlocalization

Deformation to the normal bundle

Let $i : Z \hookrightarrow X$ be a closed immersion of smooth schemes, defined by an ideal sheaf $\mathcal{I} \subseteq \mathcal{O}_X$. Write $T_Z X$ for the normal bundle and $T^*_Z X$ for the conormal bundle. The deformation space has the two descriptions

$$\begin{aligned} A_Z(X) &= \mathrm{Spec}_{\mathcal{O}_X}\big(\mathcal{O}_X[u, \mathcal{I}u^{-1}]\big) \\ &= \mathrm{Bl}_{Z\times\{0\}}(X \times \mathbf{A}^1) \smallsetminus \big\{\text{strict transform of } X \times \{0\}\big\}. \end{aligned}$$

It maps to X, and the element u defines a smooth morphism $A_Z(X) \to \mathbf{A}^1$. We have the following Cartesian diagrams:

$$\begin{array}{ccccc} Z & \hookrightarrow & Z \times \mathbf{A}^1 & \hookleftarrow & Z \times \mathbf{G}_m \\ \downarrow{\scriptstyle 0} & & \downarrow & & \downarrow{\scriptstyle i\times 1} \\ T_Z X & \hookrightarrow & A_Z(X) & \hookleftarrow & X \times \mathbf{G}_m \xrightarrow{p_1} X \\ \downarrow & & \downarrow{\scriptstyle u} & & \downarrow \\ \{0\} & \hookrightarrow & \mathbf{A}^1 & \hookleftarrow & \mathbf{G}_m \end{array}$$

Here $0 : Z \to T_Z X$ is the zero section. Let η_0 be the generic point of the henselization of $\mathbf{A}^1$ at 0. The specialization functor

$$\nu_i = \nu_{Z/X} : D(X) \longrightarrow D^b_c(T_Z X \times_k \eta_0, \Lambda)$$

is defined by $\nu_{Z/X} = R\Psi \circ p_1^*$, where $R\Psi$ is the nearby cycles functor. Here $D^b_c(U \times_k \eta_0, \Lambda)$ denotes the derived category of constructible sheaves on $U_{\bar{k}}$ with continuous action of

$$G_{\eta_0} = \mathrm{Gal}(\bar{\eta}_0/\eta_0)$$

compatible with the action of G_k on $U_{\bar{k}}$ via $G_{\eta_0} \twoheadrightarrow G_k$.

Apply this construction to the diagonal $\delta : X \hookrightarrow X \times X$. We get

$$\begin{array}{ccccc} TX & \hookrightarrow & A_X(X \times X) & \hookleftarrow & X \times X \times \mathbf{G}_m \\ \downarrow & & \downarrow & & \downarrow \\ \{0\} & \hookrightarrow & \mathbf{A}^1 & \hookleftarrow & \mathbf{G}_m \end{array}$$

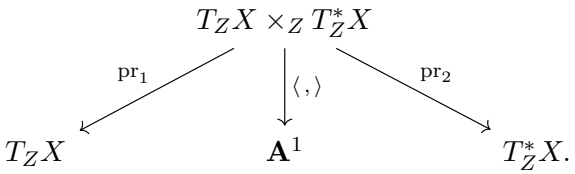
and hence

$$\nu_\delta = \nu_{X/(X\times X)} : D^b_c(X \times X, \Lambda) \longrightarrow D^b_c(TX \times_k \eta_0, \Lambda).$$

Fix a nontrivial character $\psi : \mathbb{F}_p \to \Lambda^\times$. Let $\mathcal{L}_\psi$ be the associated Artin–Schreier sheaf on $\mathbf{A}^1$, obtained from the étale cover

$$\mathbf{A}^1 \longrightarrow \mathbf{A}^1, \qquad x \longmapsto t = x^p - x.$$

Consider the two projections and the canonical pairing



The (naive) Fourier transform is

$$\begin{aligned}\mathsf{F} = \mathsf{F}_\psi = \mathsf{F}_{\psi, T_ZX/Z} :\ D_c^b(T_ZX, \Lambda) &\longrightarrow D_c^b(T_Z^*X, \Lambda),\\ \mathcal{F} &\longmapsto \mathrm{pr}_{2!}\big(\mathrm{pr}_1^*\mathcal{F} \otimes \langle\,,\,\rangle^*\mathcal{L}_\psi\big).\end{aligned}$$

The transform F_ψ depends on ψ. One can also use Wang's Fourier transform.

For $\mathcal{F} \in D(X)$, we define its *microlocalization* along $i : Z \hookrightarrow X$ to be the Fourier transform of its specialization:

$$\begin{aligned}\mu_{Z/X} :\ D(X) &\longrightarrow D(T_Z^*X \times_k \eta_0),\\ \mu_{Z/X}(\mathcal{F}) &= \mathsf{F}_\psi\big(\nu_{Z/X}(\mathcal{F})\big).\end{aligned}$$

Lemma. For the zero section $0 : Z \to T_ZX$,

$$0^*\nu_{Z/X}(\mathcal{F}) \xrightarrow{\sim} i^*\mathcal{F}, \qquad i^!\mathcal{F} \xrightarrow{\sim} 0^!\nu_{Z/X}(\mathcal{F}).$$

The complex $\nu_{Z/X}(\mathcal{F})$ on $(T_ZX)_{\bar{k}}$ is *monodromic*. For every geometric point $x : \operatorname{Spec}\Omega \to (T_ZX)_{\bar{k}}$ and its orbit map

$$\omega_x : \mathbf{G}_{m,\Omega} \longrightarrow (T_ZX)_{\bar{k}}, \qquad t \longmapsto tx,$$

the cohomology sheaves of $\omega_x^*\nu_{Z/X}(\mathcal{F})$ are locally constant and tamely ramified at 0 and ∞.

Descent of the support. For a k-scheme T and $\mathcal{F} \in D_c^b(T \times_k \eta_0, \Lambda)$, $\operatorname{supp}(\mathcal{F}) \subseteq T_{\bar{k}}$ is G_k-stable. By Galois descent for closed subsets, there is a unique closed subset $\operatorname{supp}_k(\mathcal{F}) \subseteq T$ whose inverse image under $T_{\bar{k}} \to T$ is $\operatorname{supp}(\mathcal{F})$:

$$\big(\operatorname{supp}_k(\mathcal{F})\big)_{\bar{k}} = \operatorname{supp}(\mathcal{F}) = \overline{\{t \in T_{\bar{k}} \mid \mathcal{F}_{\bar{t}} \not\simeq 0\}}.$$

Microlocalization of singular supports

Along the diagonal $\delta : X \hookrightarrow X \times X$, write

$$\mathcal{H}(\mathcal{F}, \mathcal{G}) = R\mathcal{H}om\big(\mathrm{pr}_1^*\mathcal{F}, \mathrm{pr}_2^!\mathcal{G}\big).$$

Then

$$\begin{aligned}\nu\mathcal{H}om(\mathcal{F}, \mathcal{G}) &= \nu_{X/(X\times X)}\mathcal{H}(\mathcal{F}, \mathcal{G}) && \text{on } TX,\\ \mu\mathcal{H}om(\mathcal{F}, \mathcal{G}) &= \mathsf{F}_\psi\big(\nu\mathcal{H}om(\mathcal{F}, \mathcal{G})\big) && \text{on } T^*X.\end{aligned}$$

We define

$$\mathrm{SS}_\mu(\mathcal{F}) := \operatorname{supp}_k \mu\mathcal{H}om(\mathcal{F}, \mathcal{F}).$$

Theorem A. $\mathrm{SS}_\mu(\mathcal{F}) = \mathrm{SS}(\mathcal{F})$, where $\mathrm{SS}(\mathcal{F})$ is Beilinson's singular support.

As a corollary of $\mathrm{SS}_\mu(\mathcal{F}) \subseteq \mathrm{SS}(\mathcal{F})$, for a closed immersion $i : Z \hookrightarrow X$ of smooth schemes, we have

$$\operatorname{supp}_k(\mu_{Z/X}(\mathcal{F})) \subseteq \mathrm{SS}_\mu(\mathcal{F}) \cap T_Z^*X$$

Microlocalization of characteristic cycles

For $\mathcal{F} \in D(X)$, its cohomological characteristic class $C_{X/k}(\mathcal{F}) \in H^0(X, K_{X/k})$ is the following composition. Here $\delta : X \to X \times X$ is the diagonal:

$$\begin{array}{ccccc} \Lambda & \xrightarrow{1_{\mathcal{F}}} & R\mathcal{H}om(\mathcal{F},\mathcal{F}) & \xrightarrow{\simeq} & \delta^!\mathcal{H}(\mathcal{F},\mathcal{F}) \\ & & & & \uparrow\simeq \\ & & & & \delta^!\big(D_{X/k}(\mathcal{F}) \boxtimes \mathcal{F}\big) \\ & & & & {\scriptstyle \delta^! \to \delta^*}\downarrow \\ K_{X/k} & \xleftarrow{\mathrm{ev}} & D_{X/k}(\mathcal{F}) \otimes \mathcal{F} & = & \delta^*\big(D_{X/k}(\mathcal{F}) \boxtimes \mathcal{F}\big) \end{array}$$

Lu–Zheng's reformulation

$$\begin{array}{ccc} (X,\Lambda) & \longrightarrow & \mathcal{H}om_{\mathrm{Corr}}\big((X,\mathcal{F}),(X,\mathcal{F})\big) \\ & & \| \\ & & (X,\mathcal{F}) \otimes (X,\mathcal{F})^\vee \xrightarrow{\mathrm{ev}} (X,\Lambda). \end{array}$$

The class $C_{X/k}(\mathcal{F})$ is given by the pair of arrows

$$\delta_!\Lambda \longrightarrow \mathcal{H} \longrightarrow \delta_* K_{X/k}, \qquad \mathcal{H} = R\mathcal{H}om\big(\mathrm{pr}_1^*\mathcal{F}, \mathrm{pr}_2^!\mathcal{F}\big).$$

Taking specialization and then microlocalization, we obtain, with $e^\vee : T^*X \to X$,

$$\begin{aligned} 0_!\Lambda &\longrightarrow \nu\mathcal{H}om(\mathcal{F},\mathcal{F}) \longrightarrow 0_* K_{X/k}, \\ \Lambda &\longrightarrow \mu\mathcal{H}om(\mathcal{F},\mathcal{F}) \longrightarrow e^{\vee *} K_{X/k}. \end{aligned} \tag{$**$}$$

Regard these arrows over $\bar{k}$. The composition $(**)$ defines a cohomology class in

$$H^0_{(\mathrm{SS}_\mu(\mathcal{F}))_{\bar{k}}}\big(T^*X_{\bar{k}}, q^* e^{\vee *} K_{X/k}\big), \qquad q : T^*X_{\bar{k}} \longrightarrow T^*X.$$

We have $\dim \mathrm{SS}_\mu(\mathcal{F}) = \dim X$ for nonzero $\mathcal{F}$ on a smooth equidimensional X. By semi-purity we get an isomorphism

$$q^* : H^0_{\mathrm{SS}_\mu(\mathcal{F})}\big(T^*X, e^{\vee *} K_{X/k}\big) \xrightarrow{\sim} H^0_{(\mathrm{SS}_\mu(\mathcal{F}))_{\bar{k}}}\big(T^*X_{\bar{k}}, q^* e^{\vee *} K_{X/k}\big)^{G_k},$$

where the group on the left is

$$\bigoplus_{C_a} \Lambda \cdot \mathrm{cl}[C_a], \qquad C_a \text{ an irreducible component of } \mathrm{SS}_\mu(\mathcal{F}) \text{ of dimension } \dim X.$$

The composition $(**)$ therefore defines the unique class

$$\mathrm{CC}_\mu(\mathcal{F}) \in H^0_{\mathrm{SS}_\mu(\mathcal{F})}\big(T^*X, e^{\vee *} K_{X/k}\big)$$

whose pullback by q is the class of $(**)$.

Theorem B. Let $0^\vee : X \to T^*X$ be the zero section.

1. $(0^\vee)^* \mathrm{CC}_\mu(\mathcal{F}) = C_{X/k}(\mathcal{F})$.
2. $\mathrm{CC}_\mu(\mathcal{F}) = \mathrm{cl}\big(\mathrm{CC}(\mathcal{F})\big)$, where

$$\mathrm{cl} : Z_{\dim X}\big(\mathrm{SS}(\mathcal{F})\big) \longrightarrow H^0_{\mathrm{SS}_\mu(\mathcal{F})}\big(T^*X, e^{\vee *} K_{X/k}\big).$$

Theorem B = Theorem A + Milnor formula.

Consequences of (1), using (2).

Saito's conjecture. (Below $(0^\vee)^!$ is the refined Gysin pullback)

$$\mathrm{cl}_X\big((0^\vee)^!\mathrm{CC}(\mathcal{F})\big) = C_{X/k}(\mathcal{F}).$$

Yang–Zhao previously proved this under the quasi-projectivity assumption.

Index formula. If X is proper over k, then

$$\chi(X_{\bar{k}}, \mathcal{F}) = \deg\big((0^\vee)^!\mathrm{CC}(\mathcal{F})\big) \qquad \text{in } \mathbb{Z}.$$

The previous result assumed projectivity.

Conductor formula. Let $f : X \to Y$ be proper flat, with X, Y smooth and $\dim Y = 1$. Let $y \in Y$ be a closed point. If f is smooth outside X_y, then

$$-a_y(Rf_*\Lambda) = \big(\mathrm{CC}(\Lambda), df\big)_{T^*X, X_y} \qquad \text{in } \mathbb{Z}.$$

If $n = \dim X$, this is the degree of the localized Chern class, with sign:

$$(-1)^n \deg_{k(y)}\left(c^X_{n,X_y}(\Omega^1_{X/Y}) \cap [X]\right).$$

3 Idea of the proof of Theorem A

We can reduce to $X = \mathbf{P}^n = P$. Consider the incidence correspondence

$$\begin{array}{ccccc} & & Q & & \\ & \swarrow^{p} & & \searrow^{p^\vee} & \\ P & & & & P^\vee \end{array} \qquad Q = \{(x, H) \in P \times P^\vee \mid x \in H\}.$$

Starting with $\mathcal{F}$ on P, put on Q

$$\boxed{\mathcal{G} = (p^\vee)^* R(p^\vee)_* p^* \mathcal{F}[2n-2](n-1).}$$

$$Q \simeq \mathbf{P}_{\text{lines}}(T^*P) \simeq \mathbf{P}_{\text{quot}}(TP).$$

Beilinson's result. $\mathbf{P}(\mathrm{SS}(\mathcal{F}))$ is the non-ULA (non-smooth) locus of $(p : Q \to P, \mathcal{G})$. To show that $\mathrm{SS}_\mu(\mathcal{F}) \subseteq \mathrm{SS}(\mathcal{F})$, it suffices to show that $\mu\mathcal{H}om(\mathcal{F}, \mathcal{F})$ vanishes on the punctured cotangent lines parametrized by $U = Q \smallsetminus \mathbf{P}(\mathrm{SS}(\mathcal{F}))$.
The Cartesian square on the left induces the map of normal bundles on the right:

$$\begin{array}{ccc} Q \times_P Q & \xhookrightarrow{i_p} & Q \times Q \\ \downarrow & & \downarrow{\scriptstyle p\times p} \\ P & \xhookrightarrow[\delta_P]{} & P \times P \end{array} \quad \Longrightarrow \quad \begin{array}{ccc} T_{Q\times_P Q}(Q \times Q) & \xrightarrow{\Pi} & T_P(P \times P) \\ \simeq\downarrow & & \downarrow\simeq \\ (Q \times_P Q) \times_P TP & \xrightarrow[\text{proper}]{} & TP. \end{array}$$

Compare ν_{δ_P} and ν_{i_p}.

Writing $\mathcal{H}_{\mathcal{E}} = \mathcal{H}(\mathcal{E},\mathcal{E})$, we have a retraction

$$\nu_{\delta_P}(\mathcal{H}_{\mathcal{F}}) \longrightarrow \Pi_*\nu_{i_p}(\mathcal{H}_{\mathcal{G}}) \longrightarrow \nu_{\delta_P}(\mathcal{H}_{\mathcal{F}}).$$

Thus $\mu\mathcal{H}om(\mathcal{F},\mathcal{F})$ is a retract of

$$\mathsf{F}_\psi\big(\Pi_*\nu_{i_p}(\mathcal{H}_{\mathcal{G}})\big) \simeq (\Pi^\vee)_*\mathsf{F}_\psi\big(\nu_{i_p}(\mathcal{H}_{\mathcal{G}})\big).$$

Estimate the support

$$\Sigma := \mathrm{supp}_k\big(\mathsf{F}_\psi\nu_{i_p}(\mathcal{H}_{\mathcal{G}})\big) \ \subseteq\ (Q\times_P Q)\times_P T^*P.$$

There is a linear action of

$$T\big((Q\times Q)/(P^\vee\times P^\vee)\big)\big|_{Q\times_P Q} \quad \text{on} \quad T_{Q\times_P Q}(Q\times Q)$$

under which $\nu_{i_p}(\mathcal{H}_{\mathcal{G}})$ is invariant. Hence

$$\Sigma \ \subseteq\ \text{zero section} \ \cup\ \text{tautological line bundle on } Q \hookrightarrow Q\times_P Q.$$

But $p|_U$ is ULA with respect to $\mathcal{G}|_U$. Therefore

$$\Sigma\big|_{U\times_P U} \ \subseteq\ \text{zero section}.$$

The reverse inclusion

We have shown $\mathrm{SS}_\mu(\mathcal{F}) \subseteq \mathrm{SS}(\mathcal{F})$; for the converse, by Beilinson's test-function criterion, we only need to show that a function whose differential misses $\mathrm{SS}_\mu(\mathcal{F})$ is ULA relative to $\mathcal{F}$. The assertion is local, so on the support of $\mathcal{F}$ we may shrink to a smooth test function $f : X \to \mathbf{A}^1$ with $df(X)\cap \mathrm{SS}_\mu(\mathcal{F}) = \varnothing$. Consider $i_f : X\times_{\mathbf{A}^1} X \hookrightarrow X\times X$: its defining equation $f(x)-f(y)=0$ gives the conormal line spanned by $(df_x, -df_y)$ at (x,y).

For $\mathcal{H}_{\mathcal{F}} = D_{X/k}(\mathcal{F})\boxtimes\mathcal{F}$, the Künneth formula gives

$$\mathrm{SS}_\mu(\mathcal{H}_{\mathcal{F}}) = \mathrm{SS}_\mu(D_{X/k}(\mathcal{F}))\times \mathrm{SS}_\mu(\mathcal{F}).$$

No nonzero point of the conormal line lies in this product: its second coordinate is a nonzero multiple of df_y, and $\mathrm{SS}_\mu(\mathcal{F})$ is conical, i.e.,

$$\mathrm{SS}_\mu(\mathcal{H}_{\mathcal{F}}) \cap T^*_{X\times_{\mathbf{A}^1}X}(X\times X) \subseteq X\times_{\mathbf{A}^1} X\times_{\mathbf{A}^1} 0_{T^*\mathbf{A}^1}.$$

But $\mathrm{supp}_k\mu_{i_f}(\mathcal{H}_{\mathcal{F}}) \subseteq \mathrm{SS}_\mu(\mathcal{H}_{\mathcal{F}})\cap T^*_{X\times_{\mathbf{A}^1}X}(X\times X)$, hence $\mu_{i_f}(\mathcal{H}_{\mathcal{F}})$ is supported on the zero section, which implies that i_f is $\mathcal{H}_{\mathcal{F}}$-transversal. By the diagonal criterion, f is ULA relative to $\mathcal{F}$; this proves the reverse inclusion and hence $\mathrm{SS}_\mu(\mathcal{F}) = \mathrm{SS}(\mathcal{F})$.